Lecture 6

Synchrotron Radiation

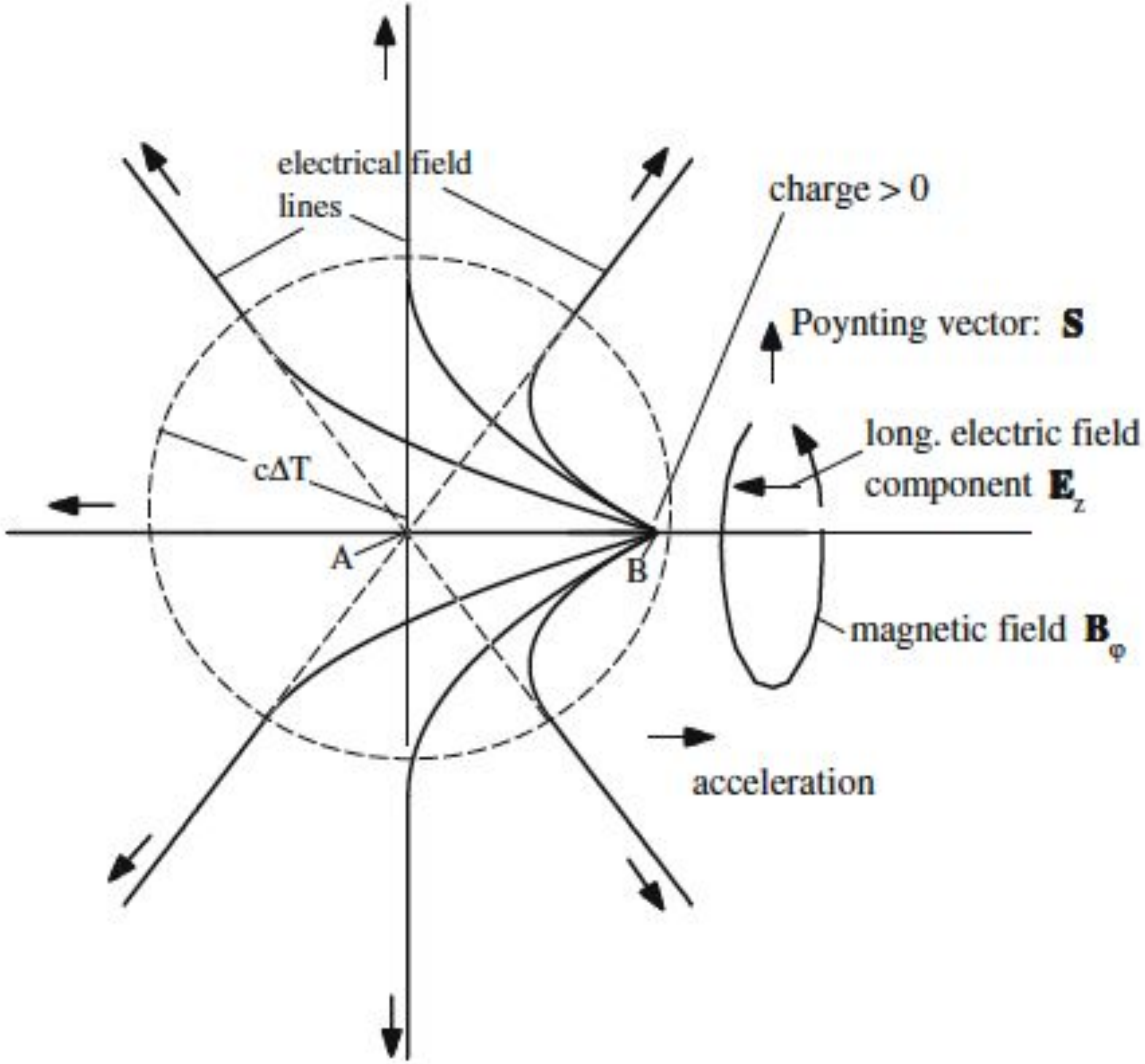
Supplemental Figures

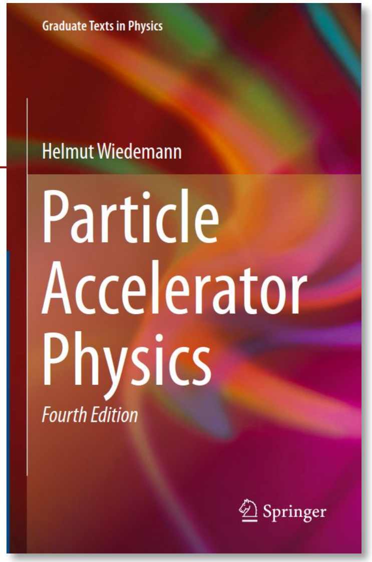
Doug Storey

7/15/2026

Field distortion due to longitudinal acceleration

Fig. 23.4 Distortion of fields due to longitudinal acceleration





Radiation distribution from an accelerating charge

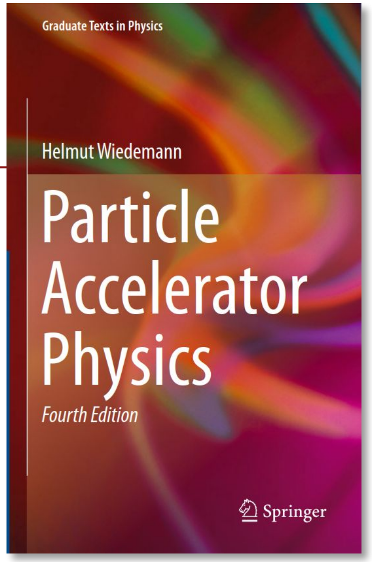
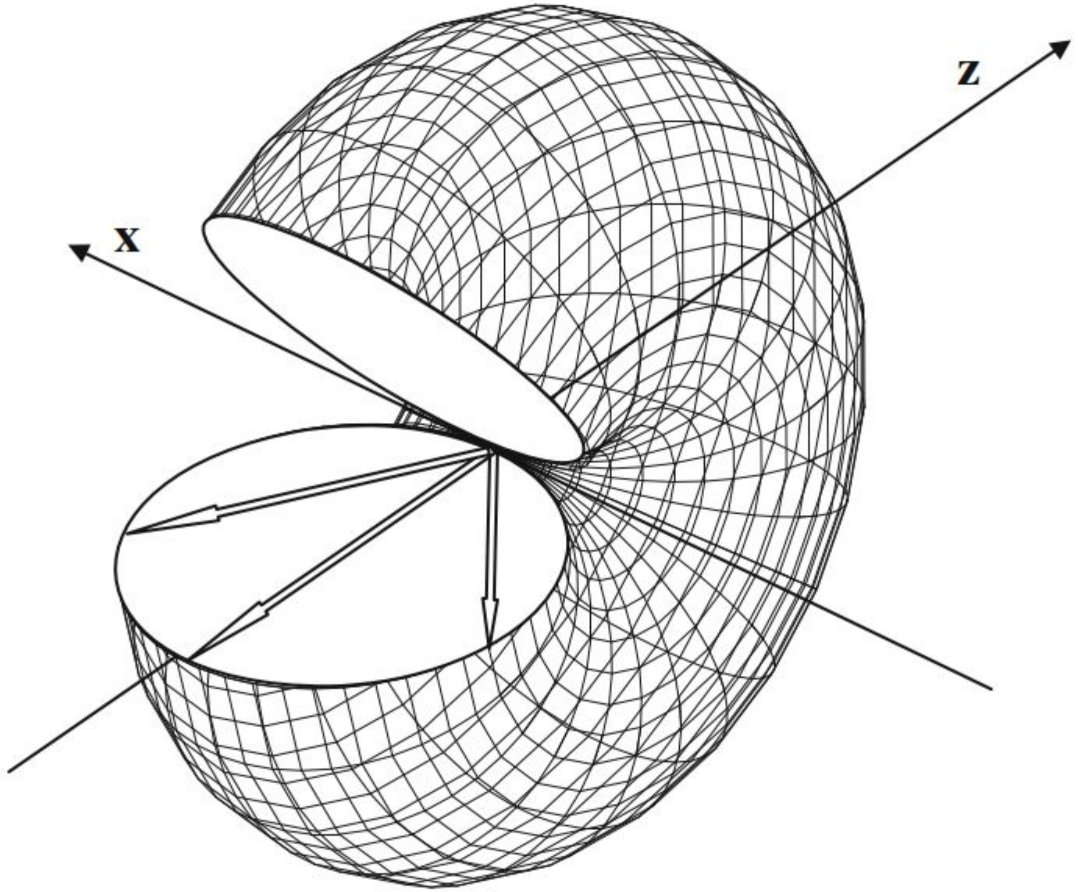
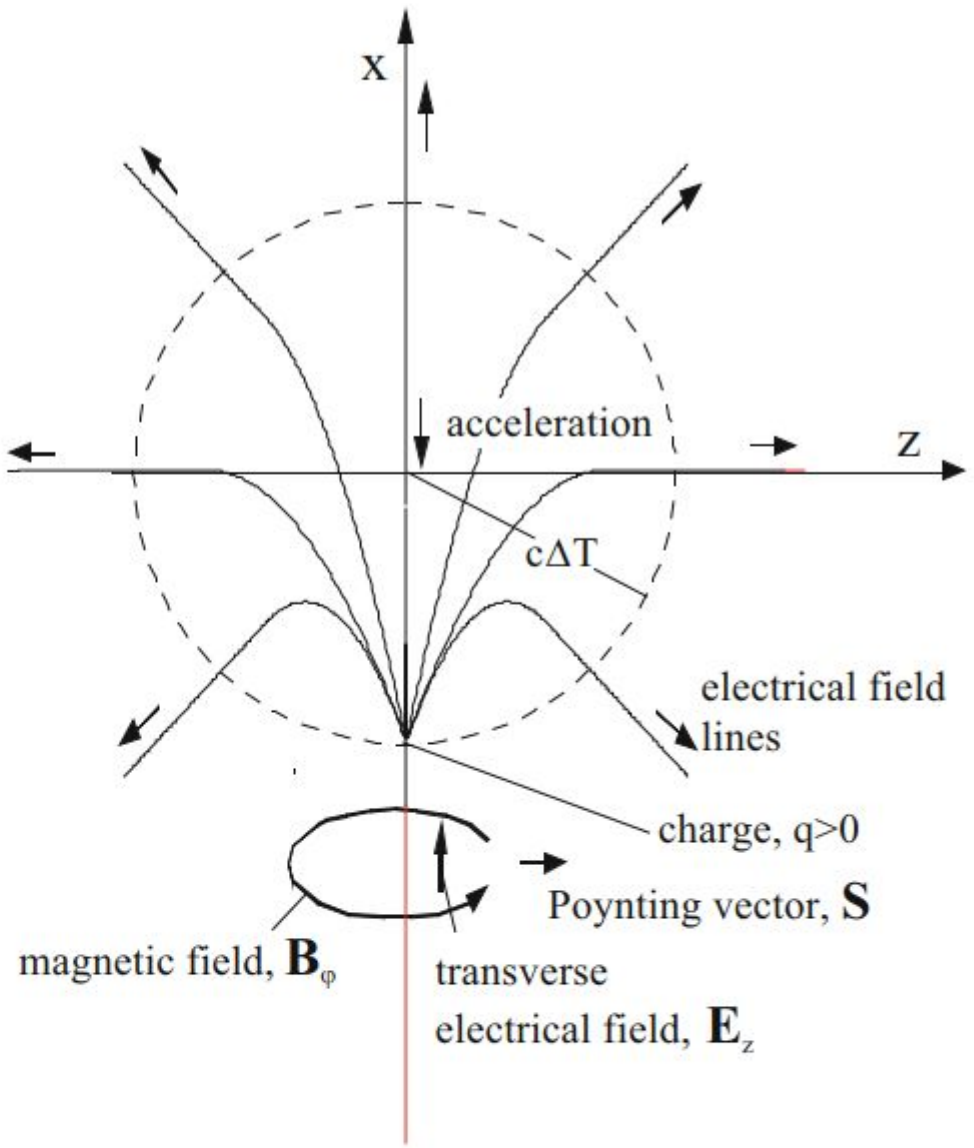


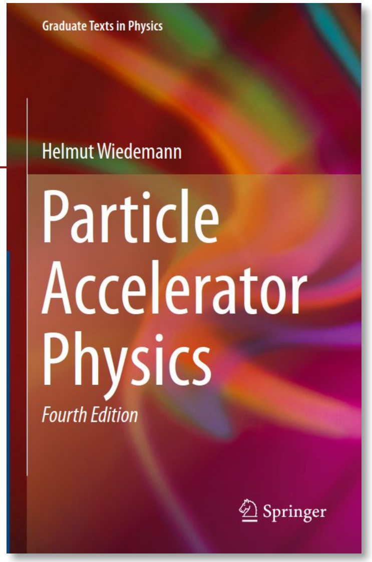
Fig. 23.5 Spatial radiation distribution in the rest frame of the radiating charge



Radiation regime – transverse acceleration

Fig. 23.6 Distortion of field lines due to transverse acceleration



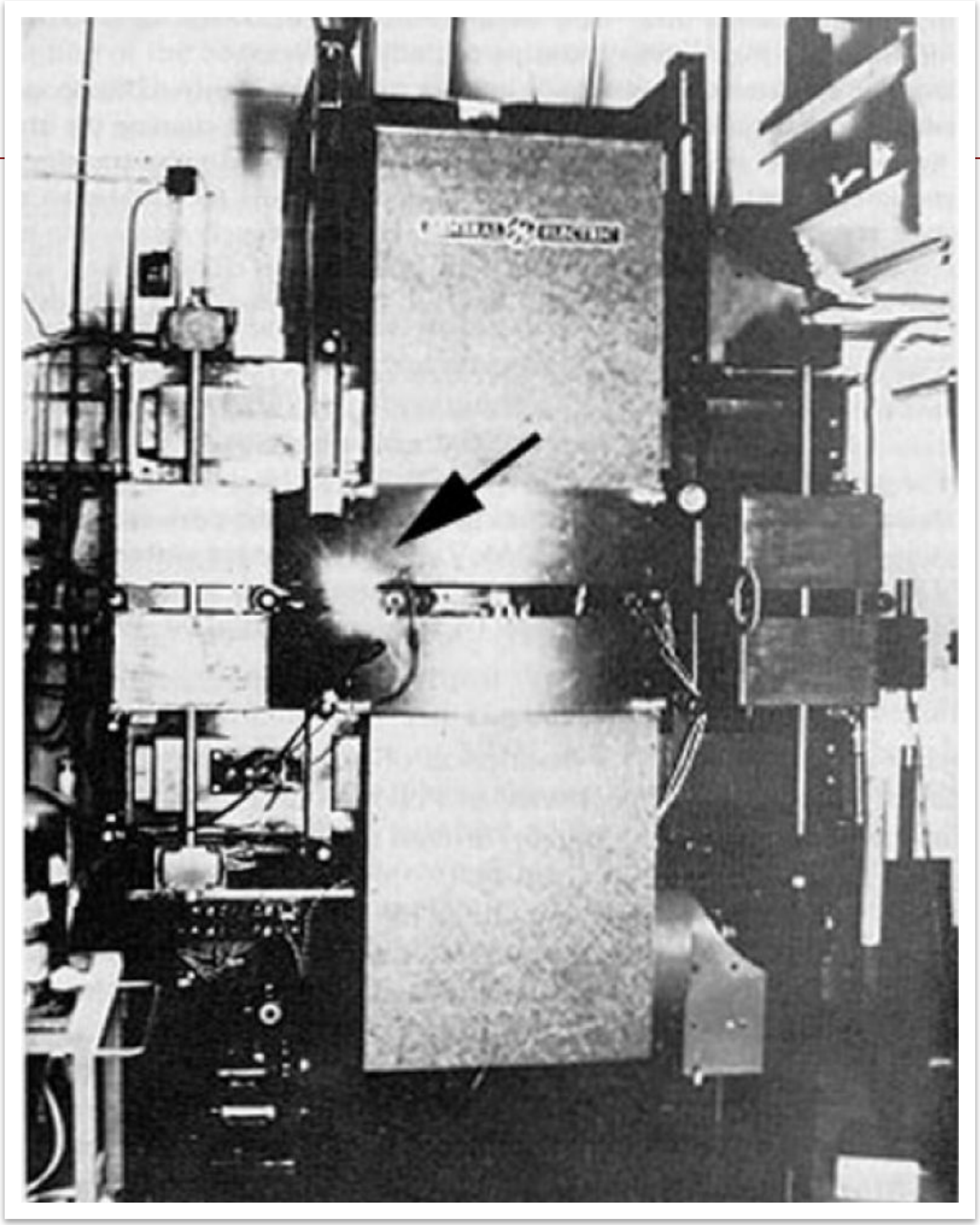


First observation of synchrotron radiation

Radiation from Electrons in a Synchrotron

F. R. Elder, A. M. Gurewitsch, R. V. Langmuir,
and H. C. Pollock
Research Laboratory, General Electric Company,
Schenectady, New York
May 7, 1947

HIGH energy electrons which are subjected to large accelerations normal to their velocity should radiate electromagnetic energy.[1-4] The radiation from electrons in a betatron or synchrotron should be emitted in a narrow cone tangent to the electron orbit, and its spectrum should extend into the visible region. This radiation has now been observed visually in the General Electric 70-Mev synchrotron.[5] This machine has an electron orbit radius of 29.3



Derivation of $\boldsymbol{E}$ of a moving charge

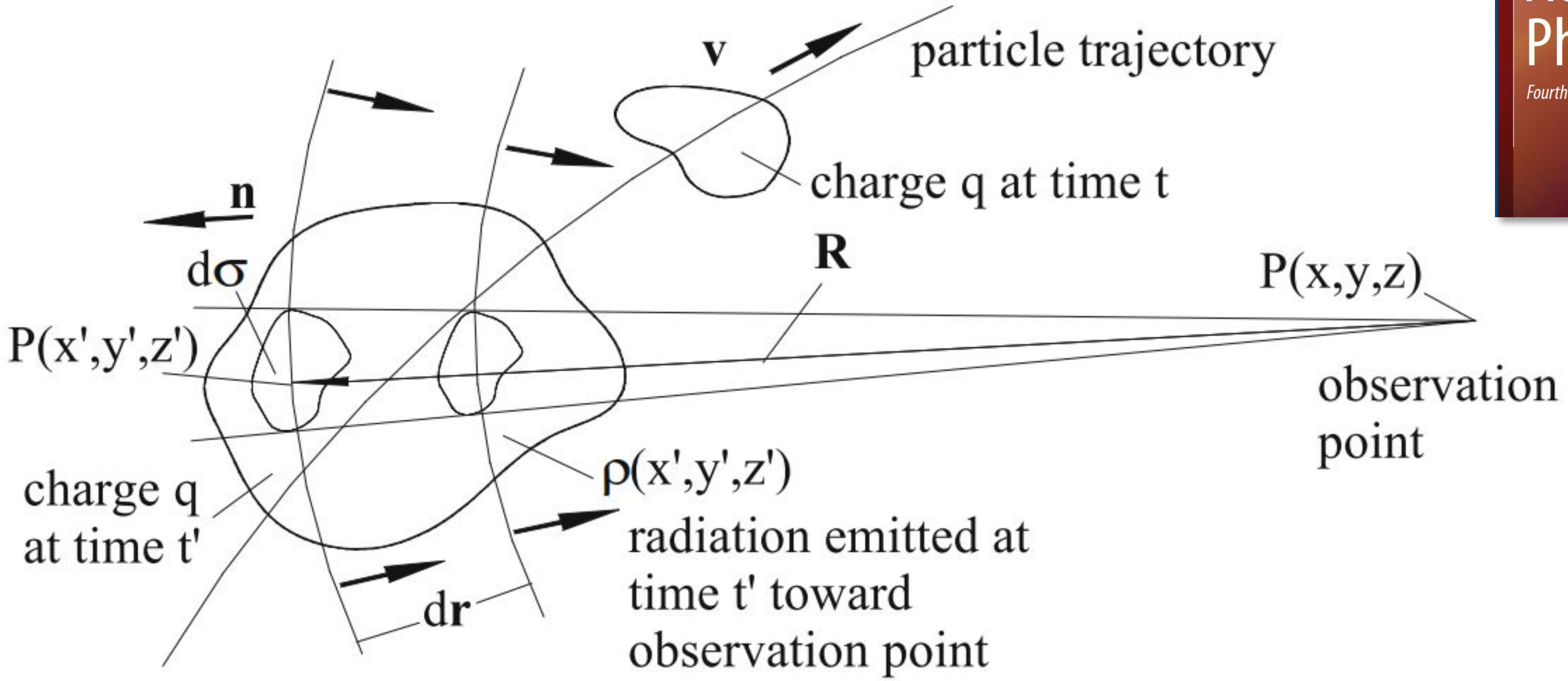
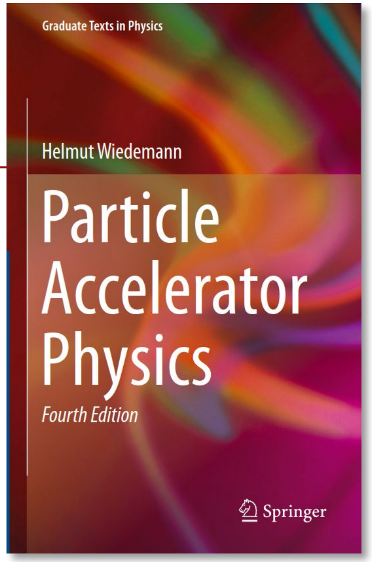
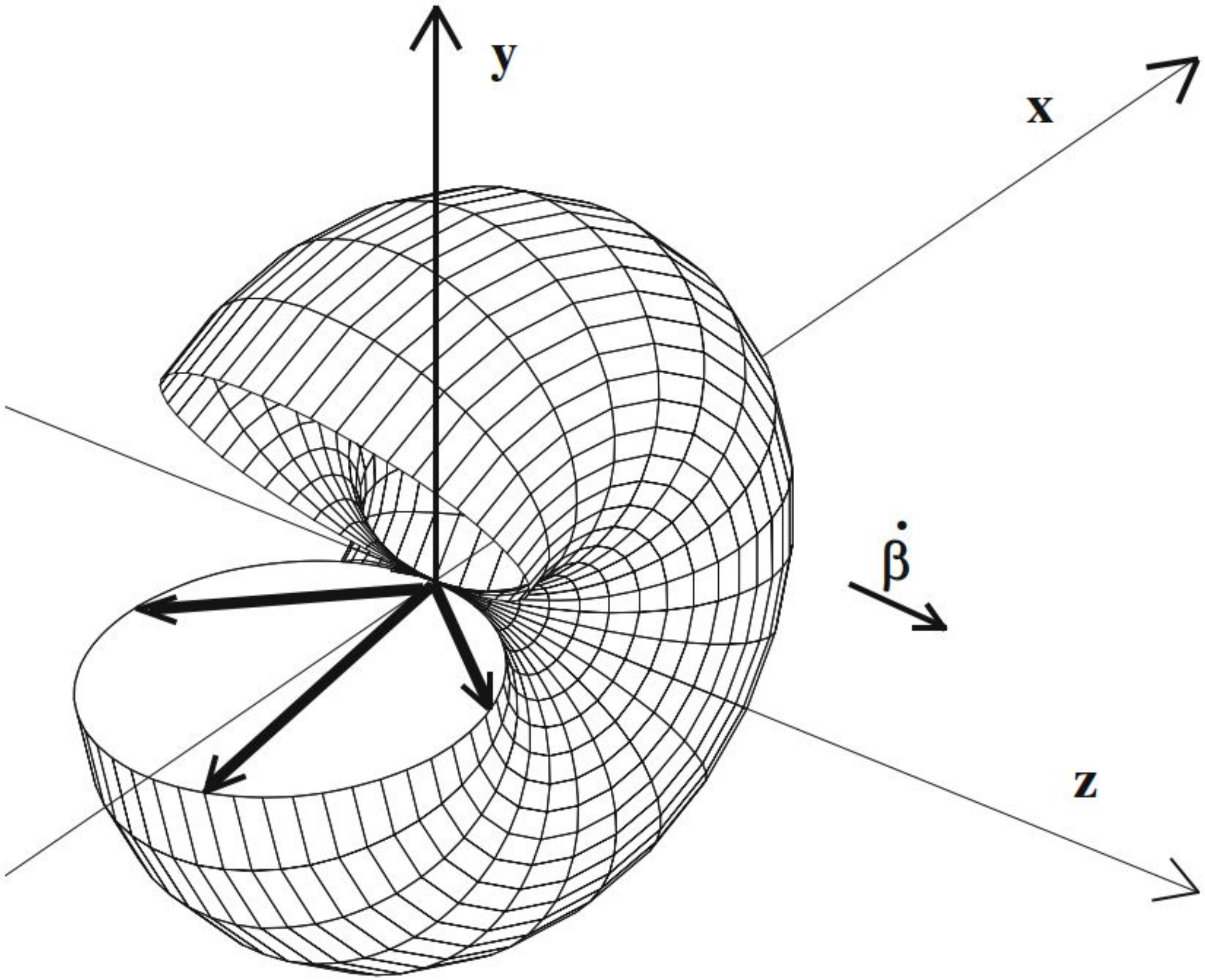


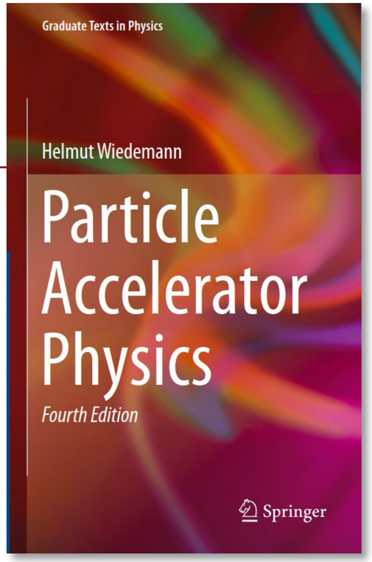
Fig. 25.1 Retarded position of a moving charge distribution



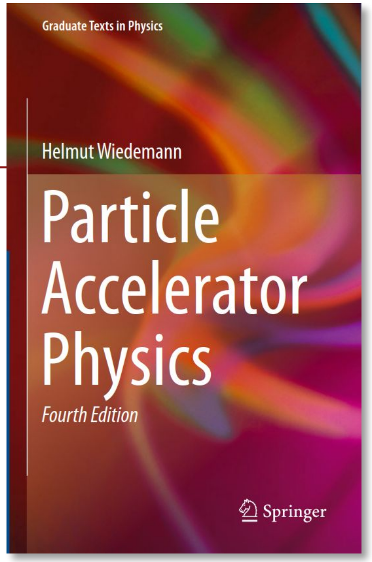
Radiation distribution in the particle's frame

Fig. 25.3 Radiation pattern in the particle frame of reference or for nonrelativistic particles in the laboratory system





Radiation distribution in the lab frame



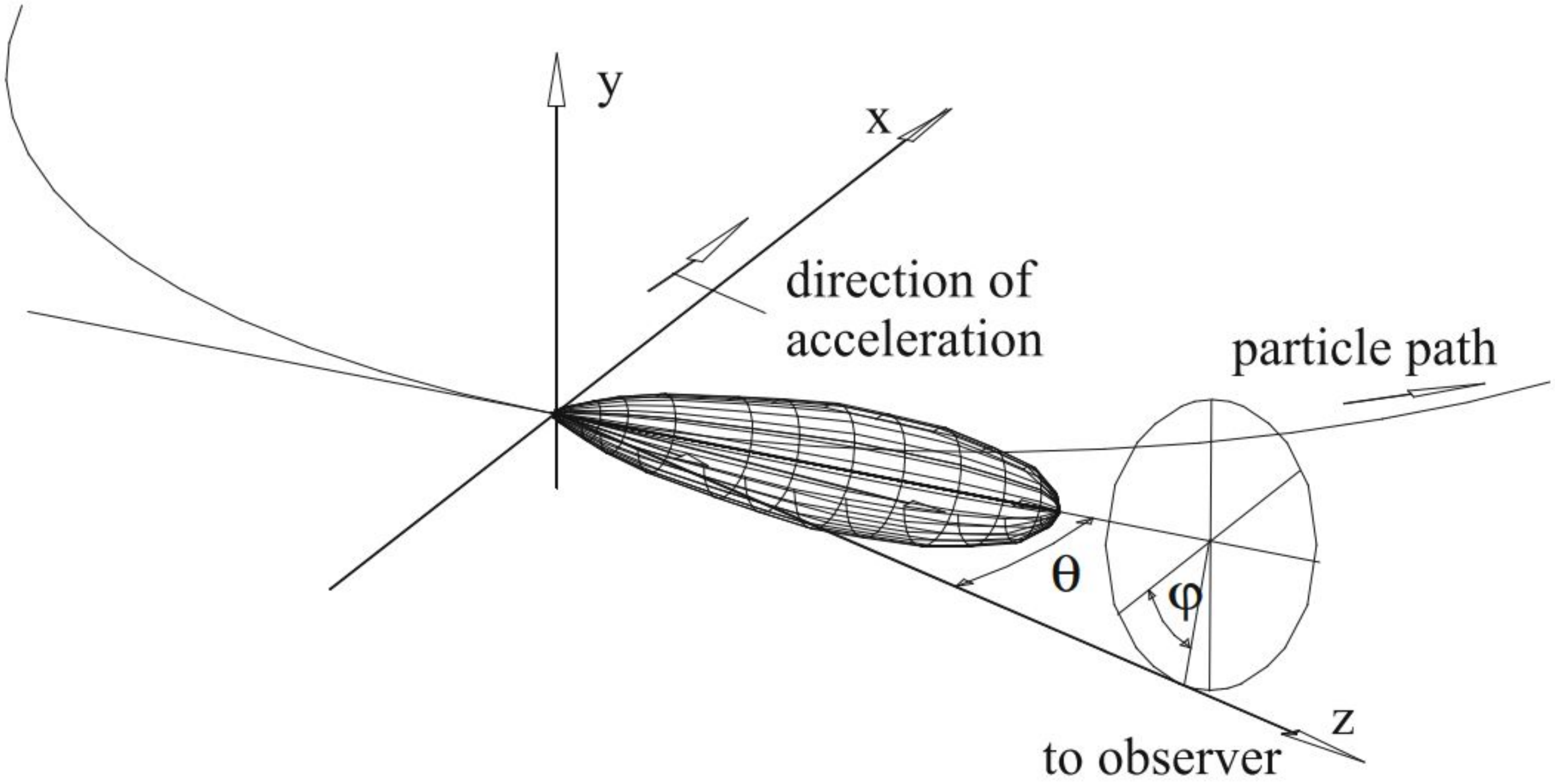
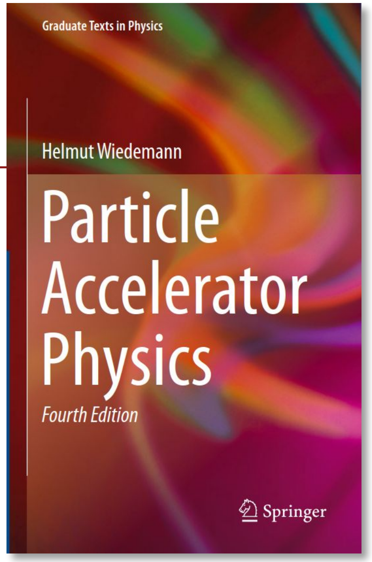


Fig. 25.4 Radiation geometry in the laboratory frame of reference for highly relativistic particles

Synchrotron radiation emitted around the ring



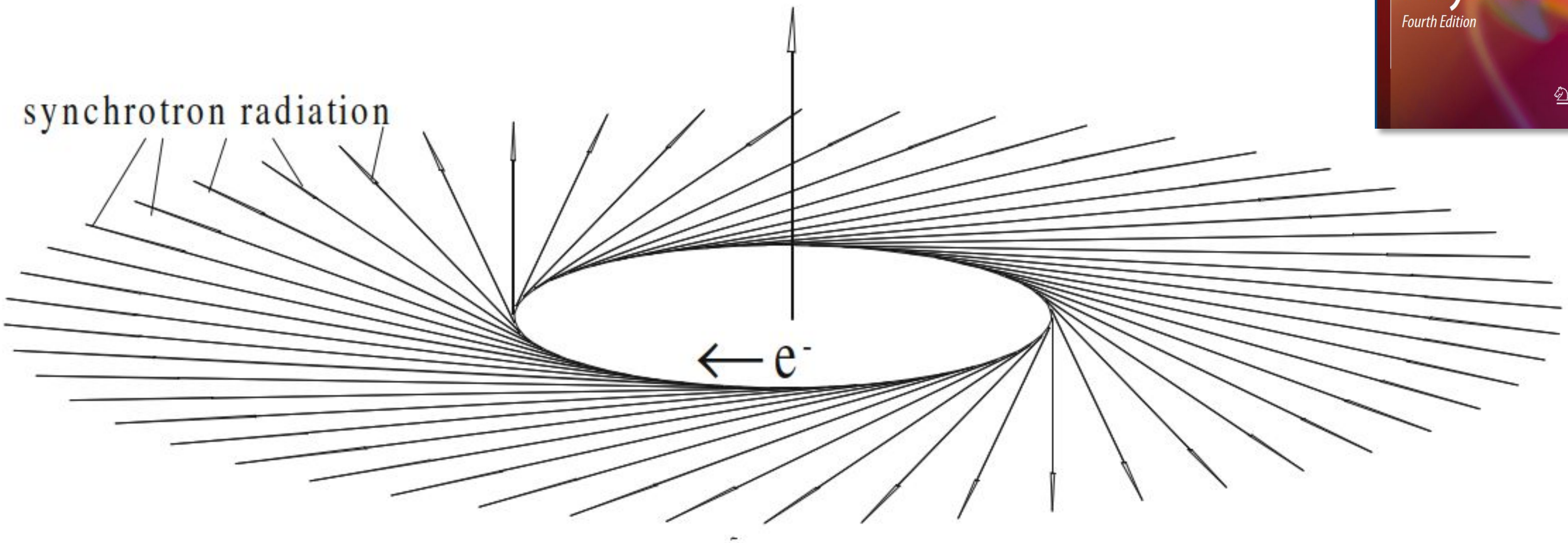


Fig. 25.6 Synchrotron radiation from a circular particle accelerator

Stable Acceleration

Synchronous Particle

$\psi_s + d\psi$ + particle

→ within separatrix

$\psi > \psi_2$ particle

→ outside of separatrix

- Off energy particle (way outside of momentum aperture)

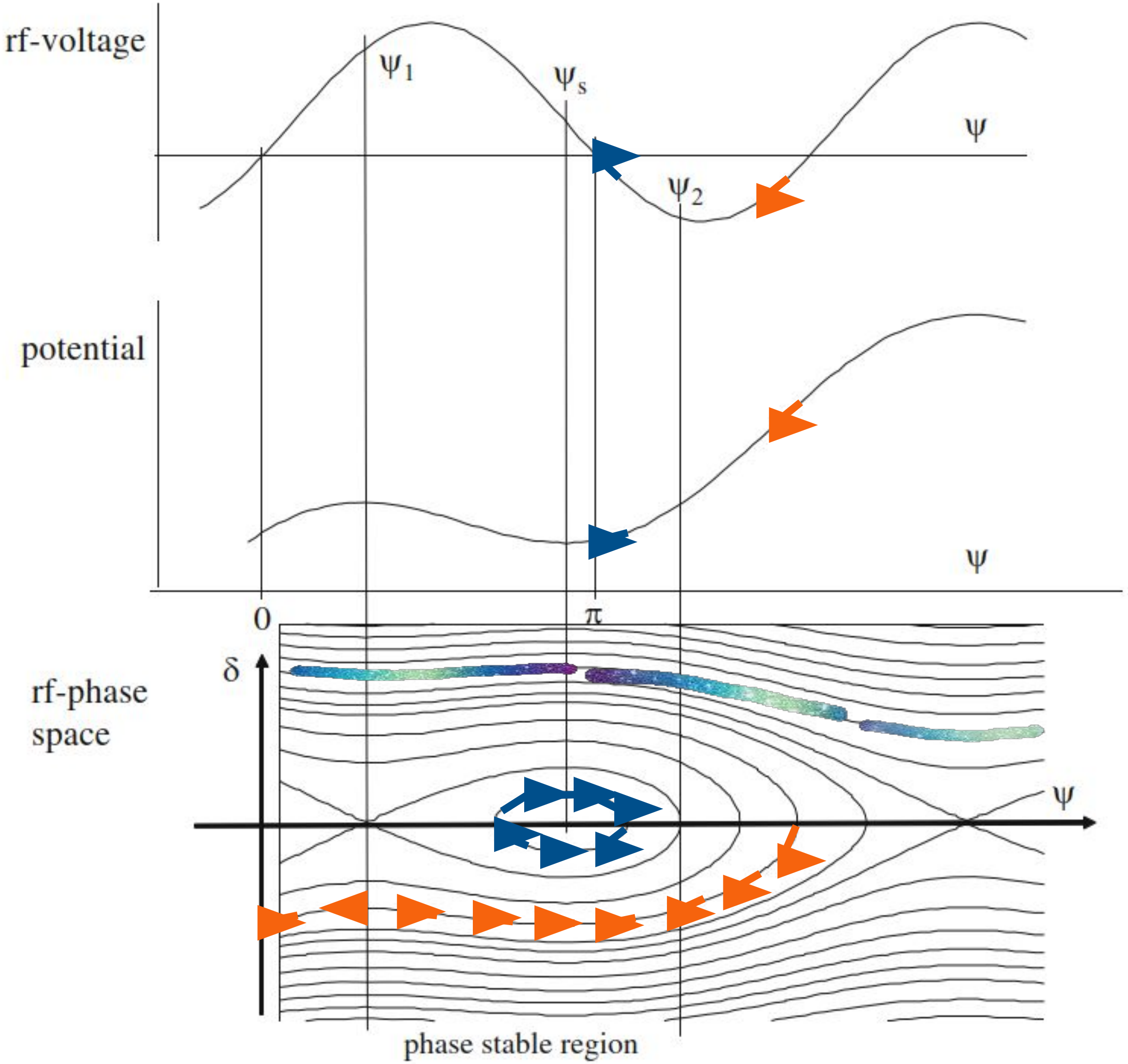


Fig. 9.9 Phase space focusing for moving rf buckets displaying the phase relationship of accelerating field, potential, and rf bucket